MIL-R-19038E
17 March 1978
SUPERSEDING
MIL-R-19038D
3 June 1969

MILITARY SPECIFICATION

RINSER-SANITIZER, CAN

This specification is approved for use by all Departments and Agencies and the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a rinser-sanitizer for cans up to 25 inches (635 mm) in outside diameter and 34 inches (864 mm) high.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

RR-C-82	-	Cans, Ash and Garbage, Corrugated, Taper-Side Zinc Coated; and Covers.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-T-60	-	Tape; Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Natick Research and Development Command, ATTN: DRDNA-TDG, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7320

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and HEat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N. W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Standard product. The rinser-sanitizer shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this specification. Modifications to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this specification shall be identifiable by all regular manufacturer's or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the Governemnt consistent with their normal commercial practices.

3.2 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial rinser-sanitizer provided the completed item complies with all provisions of this specification (see 6.3).

3.2.1 Stainless steel. Stainless steel shall conform to type 302 or 304 of ASTM A 167.

3.3 Design. The rinser-sanitizer shall consist of a circular bowl, a rotary spray nozzle, a pedestal with a base for floor mounting, foot pedal operated hot and cold (tap) water control valves, a vacuum breaker, and pipe, pipe fittings, seals, connections, and check valves required to assemble the components as an operating unit. The operation of each hot and cold pedal shall deliver 180°F (82.2°C) water and cold (tap) water, respectively, through the spray nozzle at a rate of not less than 10 gallons (37.9 liters) per minute when tested as specified in 4.3.2. The rinser-sanitizer shall be capable of cleaning cans up to 25 inches (635 mm) in outside diameter in not more than 2 minutes when tested as specified in 4.3.3. All control valves, pedals, and levers shall operate freely without binding when tested as specified in 4.3.2 and 4.3.3. The completely fabricated and assembled unit shall not leak when hydrostatically tested as specified in 4.3.1.

3.4 Construction. The rinser-sanitizer shall be constructed to permit access to all pipe, valves, connections, adjustment devices, operating pedals, and linkage without the use of special tools. Maximum height to top of spray nozzle shall be 33 inches (838 mm). Stainless steel components shall be fabricated of material conforming to 3.2.1.

3.4.1 Bowl. The bowl shall be fabricated of stainless steel, cast aluminum, carbon steel, or malleable iron at the contractor's option except that when specified (see 6.2), the bowl shall be of stainless steel. The bowl shall be dish shaped to a depth of not less than 4 inches (101 mm) and shall have an inside

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diameter which will accept can sizes up to 25 inches (635 mm) in outside diameter. A drain, of not less than 1-1/2 inch iron pipe size (i.p.s.) shall be fitted to the bowl and located to provide complete drainage. The drain pipe shall extend not less than 1-1/2 inches (38.1 mm) and not more than 6 inches (152 mm) below the bottom of the bowl and shall have a 1-1/2 inch i.p.s. male thread for drain line connection. A stainless steel spider insert of not less than four spokes shall be fastened to the bowl to support the cans during washing operations. The insert shall be positioned so as to hold the cans above the contaminated water in the bowl. A removable, perforated stainless steel screen, with vertical slots shall be fitted to the drain hole.

3.4.2 Spray nozzle. The spray nozzle shall be of the rotary type and shall be fabricated of any combination of stainless steel, brass, bronze, or aluminum. The spray nozzle shall be located in the center of the bowl, above the spider insert and shall be protected by a nozzle guard. The spray nozzle orifices shall project above the bowl rim to a height of not less than 1 inch (25 mm) or twice the inside diameter of the nozzle supply pipe, whichever is greater.

3.4.3 Pedestal and base. The pedestal and base shall be fabricated of stainless steel, cast aluminum, carbon steel, or malleable iron. The pedestal shall be attached to, and shall support the bowl and shall have a base with holes to accomodate floor anchoring bolts. The hot and cold water control valves shall be attached to the pedestal. The control valve pedal pivots shall be attached to the pedestal or to the underside of the bowl.

3.4.4 Control valves. The metal components of the hot water and the cold water control valves shall be constructed of stainless steel, brass, or bronze, or a combination thereof. The valves shall be the quick-opening, sefl-closing type equipped with springs to keep the valves closed, except when pressure is applied to the valves through the foot pedals. The valve disc shall be removable and the valves shall allow repacking of the packing glands while the valve is under pressure.

3.4.5 Foot pedals. The foot pedals shall be constructed of stainless steel, cast iron, carbon steel, malleable iron, or cast aluminum. Each pedal shall be equipped with an automatic safety lock to prevent accidental opening of the hot or cold water valves. The safety lock and the operating pedal must be depressed at the same time to actuate either the hot water or the cold water control valve.

3.4.6 Vacuum breaker. A removable, bronze body, vacuum breaker valve shall be positioned at a height of not less than 52 inches (1321 mm) from the top rim of the bowl to the bottom of the valve body.

3.4.7 Check valves. A bronze body, horizontal, swing-type check valve shall be installed in the hot water and the cold water inlet pipes before the hot water and cold water control valves.

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3.4.8 Lubrication. All friction points shall be equipped with means for lubrication in accordance with the manufacturer's standard practice.

3.5 Finish. The spray nozzle, hot and cold water control valves, check valves and vacuum breaker shall be finished in accordance with the manufacturer's standard practice. The exposed surfaces of all components fabricated of stainless steel shall have a number 4 standard polish or better. The exposed surfaces of all components fabricated of aluminum, carbon steel, malleable iron, cast iron, and all iron and steel pipe and pipe fittings shall be epoxy coated in accordance with the manufacturer's standard practice.

3.6 Instruction plate. An instruction plate shall be permanently affixed to each rinser-sanitizer in a location so as to be clearly visible to the operator. The instruction plate shall include necessary warnings and cautions and describe any procedures required for safe and efficient operation and servicing of the unit. The instruction plate shall be fabricated of corrision-resistant metal.

3.7 Identification marking. Each rinser-sanitizer shall be permanently and legibly marked with the manufacturer's model, name, trade name, or trademark readily identifiable of such known character as to be identifiable with said manufacturer.

3.8 Workmanship. The finished rinser-sanitizer shall be complete and clean, and free of scratches, dents, breaks, sharp edges and corners, deformities, burrs and slivers. All pipe connections shall be drawn tight and shall be free of crossed or stripped threads. Pipe thread sealing compound shall be removed from exposed threads.

3.8.1 Threaded fasteners. All threaded fasteners shall be drawn tight and shall be free of crossed or stripped threads.

3.8.2 Rivets. Rivets, if used, shall be tight, and rivet heads and peened ends shall be round and in full contact with the surfaces of the joined components. There shall be no empty rivet holes.

3.8.3 Welding and brazing. Welding and brazing, if used, shall be sound and free of burn holes, cracks, fractures, undercutting, excessive metal buildup, and splatter. All scale and flux shall be removed from the welds and brazes and the joints shall be finished to blend with the adjacent metal surfaces.

3.8.4 Castings. Castings shall be sound and free from blowholes, shrinkage, porosity, warping, or other defects which might render the castings unsound for use.

3.8.5 Epoxy coating. Epoxy coating shall be clean and free of imbedded foreign particles, areas of no coating, excessive buildup, flakes, blisters, peeling, chipping, scratches, smudges, or smears.

3.9 Contract data requirements. Contract data requirements shall be as specified (see 6.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. The lot shall consist of all rinser-sanitizers of one type offered for inspection at one time. The sample unit shall be one completely assembled rinser-sanitizer.

4.2.2.1 Visual examination. The rinser-sanitizer shall be examined for defects in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Standard product	Item not in accordance with manufacturers' standard product except for changes necessary to comply with specification requirements	X	
Finish (general)	Dirty, scratches		X
Spray nozzle, hot and cold water control valves, check, valves, vacuum breaker valve, drain screen	Not finished in accordance with manufacturers' standard practice	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Stainless steel components (other than above)	Not a number 4 polish or better	X	
Castings (if used)	Blowhole, porosity, shrinkage, cracks, warpage	X	
All other metal components	Not epoxy coated, areas of no coating, flaking, blisters, chipping, peeling	X	
	Imbedded foreign particles, excessive buildup, smudges, smears		X
Construction and workmanship (general)	Any component not of materials specified (where visual determination can be made)	X	
	Any component missing or not as specified	X	
	Dents, breaks, sharp edges, sharp corners, deformites, burrs, slivers	X	
Pipe and pipe fittings	Not drawn tight, crossed, or stripped threads	X	
	Pipe sealing compound not removed from exposed threads		X
Threaded fasteners	Not drawn tight		X
	Crossed or stripped threads	X	
Rivets (if used)	Loose, not in full contact with surfaces of joined components	X	
	Empty rivet holes		X
	Rivet heads not round		X
Welds and brazing (if used)	Burn holes, cracks, fractures, undercutting	X	
	Excessive metal buildup; splatter, scale, and flux deposits not removed		X
	Welds not ground to blend with adjacent metal surfaces		X

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship (general) (cont'd)			
Service access	Pipes, valves, connections, adjustment devices, operating pedals or linkage, not accessible for service, or requires special tools	X	
Lubrication	No provisions for lubrication at friction points	X	
Marking for identification	Missing, imcomplete, illegible		
Instruction plate	Missing; warnings and cautions not included	X	
	Not visible from operations position		X

4.2.2.2 Dimensional examination. Examination shall be made to determine compliance with dimensions specified. Any noncompliance shall be considered a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.2.3 End item testing. One rinser-sanitizer from the lot shall be tested as specified in 4.3.1, 4.3.2 and 4.3.3. Any nonconformance with specified requirements shall be cause for rejection of the lot.

4.2.3 Packaging inspection. An examination shall be made to determine that preservation-packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be as specified in table II. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 2.5 defects, expressed in terms of defects per hundred units.

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TABLE II. Examination of packaging

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Workmanship	Blocking and bracing inadequate, improper, or missing
Materials	Component missing, damaged, or otherwise defective
Contents	Number of rinser-sanitizer units is more or less than required

4.2.3.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-601 or PPP-B-621, examination for defects in closure and strapping shall be in accordance with the applicable container specification.

4.3 Test methods.

4.3.1 Hydrostatic pressure. The spray nozzle shall be replaced by a hydrostatic pressure gage and hydrostatic pressure shall be applied to the entire pipe system, through all valves, until the pressure is stabilized at 180 psi (1241046 Pa) on the gage. The gage shall be observed for not less than 1 minute to determine compliance with the nonleaking and failure requirements of 3.3. Any visible evidence of pressure drop on the pressure gage shall constitute failure of the test.

4.3.2 Temperature and volume. The rinser-sanitizer shall be bolted to the floor and connected to a cold water supply line and a 180°F (82.2°C) hot water supply line, each capable of delivering minimum pressure of 30 psig (206843 Pa). A thermometer, certified accurate to plus or minus 5°F at 180°F (± 3°C at 82.2°C) shall be installed in the hot water line at any convenient point before the spray nozzle. A can shall be placed in washing position. The drain pipe shall be directed into a container capable of holding not less than 20 gallons (75.5 liters) of water. The hot water control pedal shall be depressed to allow hot water to preheat the hot water system until 18-°F (82.2°C) is registered on the thermometer. The drainage container shall then be emptied and repositioned under the drain. The hot water control pedal shall then be depressed for not more than two minutes. During this period the thermometer shall be observed to determine that the temperature does not drop below 180°F (82.2°C) to determine compliance with the temperature requirements of 3.3. The water in the drainage container shall then be measured to determine compliance with the 10 gallon (37.9 liters) per minute volume requirement of 3.3. Cold (tap) water volume test shall be conducted the same as hot water test except temperature testing is not applicable. Failure to comply with the volume and temperature requirements of 3.3 shall constitute failure of this test. Also, any evidence of valves, levers or pedals to function as specified in 3.3 shall constitute failure of this test.

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4.3.3 Operation. A can, conforming to the requirements of size 32 gallon (121 liters) of RR-C-82 shall be coated over all interior surfaces with a mixture of either lard and coloring, or solid shirtening and coloring and allowed to set for a period of not less than 24 hours in an ambient temperature of 70°F ± 5°F (21.1°C ± 3°C). The can shall then be placed on a rinser-sanitizer and cleaned for 2 minutes in accordance with the manufacturers' instructions. The interior surfaces of the can shall then be examined visually to determine compliance with the cleaning requirements of 3.3. Any visible evidence of shortening shall constitute failure of this test. Also, any evidence of valves, levers or pedals to function as specified in 3.3 shall constitute failure of this test.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. Protruding and detachable components, such as the vacuum breaker valve and associated piping, shall be removed from each rinser-sanitizer in order to prevent damage to the components or to reduce the cube of the pack.

5.1.1.2 Sealing of openings. All threaded openings shall be sealed with threaded metal or plastic caps or plugs. All nonthreaded openings shall be sealed with tape conforming to PPP-T-60.

5.1.1.3 Detached components. Detached components shall be packaged in a sung-fitting fiberboard box conforming to style optional, grade W5c or W5s of PPP-B-636. Each box shall be closed and waterproofed in accordance with method V as specified in the appendix of the box specification.

5.1.2 Level C. The rinser-sanitizer shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The contractor may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Each rinser-sanitizer, packaged as specified in 5.1, shall be packed in a shipping container conforming to overseas type, type 3 load of PPP-B-601 or class 2, style 2 or 4, type 3 load of PPP-B-621. The fiberboard box containing detached components and technical publications shall be secured in an unused portion of the shipping container. Lumber blocking and bracing shall be used to immobilize the contents within the shipping container. The blocking and bracing shall not depend on the end-grain nailing alone, but supplemental cleats shall be provided to reinforce and prevent splitting of the blocking and bracing. Closure and strapping shall be in accordance with the appendix of the applicable container specification.

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5.2.2 Level B. Each rinser-sanitizer, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except that the shipping container shall conform to domestic type, style A or B, type 3 load of PPP-B-601, or class 1, style 2 or 4, type 3 load of PPP-B-621.

5.2.3 Level C. Rinser-sanitizers, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the contract, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The rinser-sanitizers are intended for rinsing, washing, and sanitizing waste and garbage cans and may also be used for preheating other containers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When stainless steel bowl is required (see 3.4.1).
(c) Data required (see 3.9).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification (see 3.2).

6.4 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.5 Supersession data. The rinser-sanitizer covered by this document is identical to the type I and type II units formerly covered by MIL-R-19038D. The steam-water mixing valve formerly furnished with the type I unit is no longer a component to be supplied with the rinser-sanitizer.

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - YD
Air Force - 99

Preparing activity:

Army - GL

Project No. 7320-0380

Review activities:

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